

NN-Adaptive Bayesian Quantum Estimation

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Introduction

The aim of this project was to learn *when* to measure a qubit so that its frequency is learned as fast as physically possible. The work implements adaptive Bayesian experimental design for single-qubit Ramsey frequency estimation: a sequential Monte Carlo filter maintains a posterior over the unknown Larmor frequency, a policy reads that posterior and chooses the interrogation time for the next shot, the measurement outcome sharpens the posterior, and the cycle repeats.

Two policies are trained and compared: **trust-region policy optimisation** (TRPO) and the **cross-entropy method** (CEM). Both were evaluated over $1.00\text{e}+04$ held-out frequencies. **They land in a statistical tie** — $2.17\text{e}-03$ and $2.14\text{e}-03$ Bayes risk respectively, a 2% difference — despite completely different optimisers. This report explains the experiment and the estimation problem, describes how each of the two methods works, and compares their evaluation plots.

Table 1: Abbreviations used in this report.

Full name	Abbreviation
sequential Monte Carlo	SMC
trust-region policy optimisation	TRPO
cross-entropy method	CEM
effective sample size	ESS
reinforcement learning	RL
multilayer perceptron	MLP
Kullback–Leibler divergence	KL

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1 The physics

1.1 The qubit and its state

The system is a single qubit, described by a density matrix ρ . The physically relevant feature of that state is the off-diagonal **coherence** ρ_{01} , which is what carries the phase accumulated from the unknown frequency. The qubit is prepared in the equal superposition $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, so it starts on the equator of the Bloch sphere with maximal coherence and equal populations. The parameter being estimated is the Larmor frequency ω — a detuning — which sets how fast that phase winds. Dephasing drives the coherence toward zero without changing the populations, and it is precisely that loss of coherence which removes information from the state.

1.2 The experiment

The experiment is **Ramsey interferometry** in the presence of pure dephasing. Each shot runs the same sequence:

1. Prepare $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.
2. Let the qubit precess freely under $H = \frac{\omega}{2}\sigma_z$ for a time t . This time is the **control variable** — the one thing the experimenter chooses.
3. Apply a $\pi/2$ read-out pulse and measure in the computational basis.
4. Record the binary outcome $d \in \{0, 1\}$.

The qubit is not isolated. Transverse noise destroys the coherence at a rate set by the coherence time $T_2 = 1.00e + 01$,

$$\rho_{01}(t) = \rho_{01}(0) e^{-t/T_2} e^{-i\omega t}. \quad (1)$$

Because ω and t only ever appear as the product ωt , and T_2 shares the units of t , units are arbitrary throughout.

1.3 The likelihood

Solving the Lindblad pure-dephasing master equation and applying the Born rule gives the outcome probability

$$\begin{aligned} P(d = 0 \mid \omega, t) &= e^{-t/T_2} \cos^2\left(\frac{\omega t}{2}\right) + \frac{1 - e^{-t/T_2}}{2} \\ &\equiv \frac{1}{2} [1 + e^{-t/T_2} \cos(\omega t)], \end{aligned} \quad (2)$$

with $P(d = 1 \mid \omega, t) = 1 - P(d = 0 \mid \omega, t)$. The two forms are algebraically identical. The expression reads as a **decaying interference fringe**:

Table 2: The likelihood.

term	meaning
$\cos^2(\omega t/2)$	the Ramsey fringe — this is what carries information about ω
$v(t) = e^{-t/T_2}$	the visibility , or coherent fraction
$(1 - v)/2$	the fully dephased fraction, which returns a coin flip and tells you nothing

As $t \rightarrow \infty$ the visibility vanishes, $P(d = 0) \rightarrow 1/2$, and the measurement becomes pure noise. **Decoherence therefore imposes a hard ceiling on how long it is worth interrogating.** Both the simulator and the filter use this same expression, and T_2 is treated as exactly known: the model is well specified, with no model mismatch and no nuisance parameter.

1.4 Why long interrogation times stop helping

Differentiating the likelihood gives the single-shot Fisher information

$$I(\omega; t) = \frac{(\partial_\omega p_0)^2}{p_0(1 - p_0)} = \frac{t^2 v^2 \sin^2(\omega t)}{1 - v^2 \cos^2(\omega t)}, \quad v = e^{-t/T_2}. \quad (3)$$

Without decoherence ($v \rightarrow 1$) this is just $I = t^2$: information grows quadratically in interrogation time, which is why longer is better. With finite T_2 the envelope $t^2 e^{-2t/T_2}$ peaks at $t = T_2$, so the information per shot is **bounded** at $I_{\max} \approx 1.35e - 01 T_2^2$. Numerically, for $T_2 = 1.00e + 01$ the optimum sits at $t^* \approx 8.40e + 00 - 9.50e + 00$, depending on ω . Beyond T_2 the visibility collapses faster than t^2 grows, and additional interrogation time buys nothing.

1.5 The competing constraint: phase ambiguity

Fisher information is a *local* quantity, and maximising it is the wrong move while the prior is still broad. The likelihood depends on ω only through $\cos(\omega t)$, which is periodic. Over the prior support $\Delta\omega = 1.00e + 00$ the fringe wraps once $t > 2\pi \approx 6.28e + 00$: several well-separated ω then produce the same phase, the posterior goes **multimodal**, and the filter cannot resolve the aliases.

A good policy must therefore trade off three regimes:

- **small** t — unambiguous, but low information per shot ($I \sim t^2$);
- $t \approx T_2$ — maximum information per shot, but aliases a broad prior;
- $t \gg T_2$ — decohered, zero information, strictly wasted.

The resolution is to grow t *as the posterior narrows*, keeping the phase spread across the posterior at roughly one radian. An optimal adaptive policy is therefore expected to show a rising interrogation time as the posterior collapses.

2 The estimation problem

2.1 The figure of merit: Bayes risk

This is a **Bayesian** estimation problem, and the quantity everything is judged by is the **Bayes risk**,

$$r(h \mid p(\theta)) = \mathbb{E}_{D_k} [\text{tr Cov}(\theta \mid D_k)], \quad (4)$$

where h is the experiment-design policy, $D_k = (d_1, \dots, d_k)$ the measurement record, and smaller is better. It is the risk of the quadratic loss $L(\hat{\theta}_k, \theta) = \|\hat{\theta}_k - \theta\|_2^2$ under the Bayes estimator $\hat{\theta}_k(D_k) = \mathbb{E}[\theta \mid D_k]$, that is, the posterior mean. For the single-parameter, experiment-limited case implemented here, the trace is over a 1×1 covariance and the expectation is estimated by averaging over true ω values. **The Bayes risk is therefore precisely the mean final posterior variance** by which the methods are ranked below.

2.2 The reward and its telescoping identity

The reward is the **reduction in traced posterior covariance** at each step,

$$\begin{aligned} R(D_k) &= \text{tr Cov}(\theta \mid D_{k-1}) - \text{tr Cov}(\theta \mid D_k) \\ &\xrightarrow{d=1} V_{k-1} - V_k, \end{aligned} \quad (5)$$

with $V_k = \sum_i w_i (\omega_i - \bar{\omega})^2$. This is not an arbitrary choice: the negative Bayes risk is the obvious reward but is far too slow to compute inside a training loop, whereas Eq. (5) is cheap in the filter framework *and* telescopes to it. For N measurements per episode,

$$\mathbb{E}_{D_N} \left[\sum_{j=1}^N R(D_j) \right] = \text{const} - r(h \mid p(\theta)), \quad \text{const} = \text{tr Cov}(\theta). \quad (6)$$

So maximising the episode return **provably minimises the Bayes risk** — provided the return is undiscounted.

2.3 The inference layer: the particle filter

ω is a **static** parameter — it does not evolve — so the filter only ever reweights particles; the particle locations move solely during resampling. The posterior is represented by N particles $\{\omega_i\}$ with log-weights $\{\log w_i\}$, updated in the log domain,

$$\log w_i \leftarrow \log w_i + \log P(d \mid \omega_i, t) - \text{logsumexp}(\cdot), \quad (7)$$

and renormalised every step. The filter’s health is monitored by the Kish effective sample size $\text{ESS} = 1 / \sum_i w_i^2 \in [1, N]$.

Resampling fires when $\text{ESS} < 7.50e - 01 N$, an aggressive threshold, so it triggers on most steps. The scheme used throughout is **Liu–West kernel resampling** ($a = 9.80e - 01$),

$$\omega'_i = a \omega_{\text{idx}(i)} + (1 - a) \bar{\omega} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, h^2 \Sigma), \quad h^2 = 1 - a^2. \quad (8)$$

The shrinkage toward the weighted mean exactly cancels the variance added by the jitter ($a^2 \Sigma + h^2 \Sigma = \Sigma$), so the posterior’s first two moments are preserved. The jitter is what prevents particle impoverishment: a static parameter is never diffused by process noise, so plain multinomial resampling would collapse all particles onto a handful of duplicated values.

2.4 The control problem

The control problem is framed as a standard sequential decision problem:

Table 3: The control problem.

component	definition
observation	posterior mean, posterior variance, and the last $3.00e+01$ interrogation times ($3.20e+01$ inputs)
action	interrogation time $t \in [1.00e-01, 3.00e+03]$
reward	$V_{k-1} - V_k$
episode	$1.00e+02$ measurements (training) / $1.25e+02$ (evaluation), one true ω per episode

3 Parameters

A single parameter set is shared by everything, so all comparisons are apples-to-apples.

Table 4: Parameters.

symbol	value	meaning
ω (prior)	Uniform(0, 1), $\text{Var}[\omega] = 8.33e-02$	unknown Larmor frequency / detuning
T_2	$1.00e+01$	transverse coherence time
t (control)	$[1.00e-01, 3.00e+03]$	interrogation time chosen per shot
episode length	$1.00e+02$ / $1.25e+02$	measurements per episode (train / eval)
particles	$2.00e+03$	SMC particles (training and evaluation)
evaluation set	$1.00e+04$	true ω values
discount factor	$9.90e-01$	TRPO reward discount
resample threshold	$7.50e-01N$	ESS below which resampling fires
Liu–West shrinkage	$9.80e-01$	kernel parameter a
target KL	$1.00e-02$	TRPO trust-region budget

With $\omega \lesssim 1.00e+00$ and $T_2 = 1.00e+01$, roughly $\omega T_2 \lesssim 1.00e+01$ radians of phase accumulate before coherence is lost, so a few fringes are resolvable within the coherence window. Over the prior support the fringe first wraps at $t = 2\pi \approx 6.28e+00$, which sets the boundary between the aliasing-safe and aliasing-prone regimes.

4 How TRPO works

TRPO is a policy-gradient method. At each update it maximises a linear surrogate of the expected return while constraining the Kullback–Leibler divergence between the old and the new policy to a trust region. Bounding the step size is what keeps the method stable: a single update cannot move the policy arbitrarily far, which matters because the control problem is non-stationary as the posterior evolves. A separately trained value function supplies a baseline that reduces the variance of the advantage estimate used in the surrogate.

In this implementation both the policy and the value function are multilayer perceptrons with two hidden layers of $2.56\text{e}+02$ units each. The actor is an unbounded Gaussian; its sampled action is clipped to the allowed interrogation-time interval inside the environment step. The policy is trained **from scratch**, with no supervised initialisation; the training procedure and its scale are given in [Training](#).

5 How CEM works

CEM is a gradient-free evolutionary search. It maintains a Gaussian distribution over a flat parameter vector, samples a population from it, evaluates every member of the population on the task, keeps the best elite fraction, and refits the distribution to those elites. Iterating this concentrates the search distribution on high-return parameters without ever computing a gradient.

Here the parameter vector is the flattened $5.45\text{e}+02$ -dimensional weight vector of a small policy network with one hidden layer of $1.60\text{e}+01$ units and a hyperbolic-tangent activation. Its output is squashed by a sigmoid,

$$t = t_{\min} + (t_{\max} - t_{\min}) \sigma(z), \tag{9}$$

so the network can never emit an out-of-range interrogation time — the action bound is structural rather than clipped after the fact. The training procedure and its scale are given in [Training](#).

6 Training

Both policies are trained in the same environment. Each episode draws one true frequency $\omega \sim \text{Uniform}(0, 1)$ and runs for $1.00\text{e}+02$ measurements. The posterior is maintained by the particle filter with $2.00\text{e}+03$ particles, and at every step the agent observes the posterior mean, the posterior variance and the last $3.00\text{e}+01$ interrogation times ($3.20\text{e}+01$ inputs), and receives the reward $V_{k-1} - V_k$. The discount factor is $9.90\text{e}-01$.

TRPO. The policy is trained **from scratch**, with no supervised initialisation, for $1.00\text{e}+05$ episodes — $1.00\text{e}+07$ timesteps — with a target KL of $1.00\text{e}-02$. Checkpoints are written every

5.00e+04 steps, so a partially trained policy can be evaluated at any point and a crash costs at most one interval. A full run takes **2 h 24 m** on CPU.

CEM. The gradient-free search uses a population of 1.00e+03, an elite fraction of 1.00e-01 and 1.00e+02 generations, with 2.00e+03 particles and 1.00e+02-step episodes. The returned policy is the mean of the final search distribution. A full run takes **54 minutes** on CPU.

Both runs log their per-episode (TRPO) or per-generation (CEM) metrics and their checkpoints to MLflow.

7 Evaluation

The trained policies are evaluated on the NumPy reference harness, which runs at float64 one frequency at a time. Evaluation uses 1.00e+04 held-out frequencies, drawn independently of the training draw; 1.25e+02 measurements per episode; 2.00e+03 particles; Liu–West resampling; and $T_2 = 1.00e + 01$. The figure of merit is the Bayes risk of Eq. (4), which for this single-parameter, experiment-limited problem is exactly the mean final posterior variance averaged over the 1.00e+04 frequencies. Lower is better. The analysis below reports this quantity.

8 Analysis

Table 5 gives the headline numbers over the 1.00e+04 held-out frequencies. Lower Bayes risk is better.

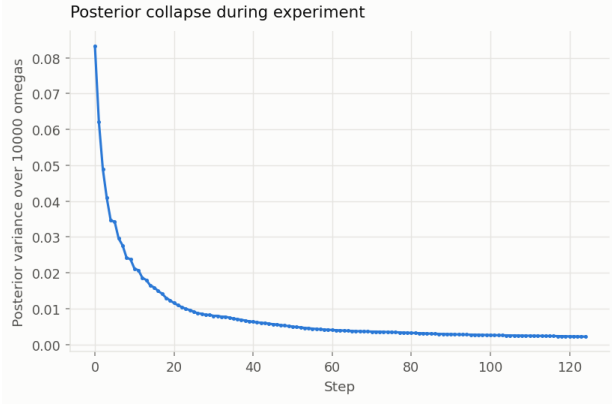
Table 5: TRPO and CEM over 1.00e+04 held-out frequencies.

quantity	TRPO	CEM
training	1.00e+05 episodes, 1.00e+07 timesteps, 2 h 24 m	1.00e+02 generations × 1.00e+03 population, 54 min
Bayes risk over 1.00e+04 ω	2.17e-03	2.14e-03
mean t	4.00	6.67
min / max t	0.10 / 9.85	2.16 / 15.60
final ESS (of 2.00e+03)	1960	1904

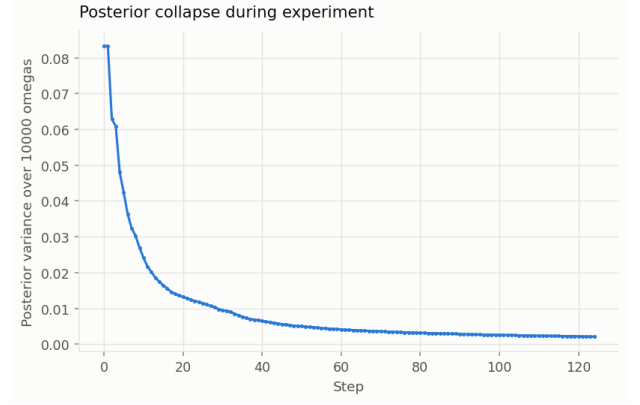
The two methods land in a **statistical tie** — 2.17e-03 against 2.14e-03, a 2% difference — despite completely different optimisers and very different chosen times: CEM settles on interrogation times roughly 1.7× longer than TRPO’s, and never goes below $t = 2.16$, where TRPO ranges down to the action floor.

8.1 Posterior averaged over omegas

Mean posterior variance across all $1.00\text{e}+04$ ω . Both policies fall from the prior's $8.33\text{e}-02$ to about $2.10\text{e}-03$ and flatten out well before the $1.25\text{e}+02$ nd measurement.



(a) TRPO



(b) CEM

Figure 1: Posterior averaged over omegas for TRPO and CEM.

9 Summary

This project implements adaptive Bayesian frequency estimation for a single qubit, with a sequential Monte Carlo filter tracking the posterior and a learned policy choosing the interrogation time at each shot. The physics and the estimation machinery are in place and behave: the likelihood, the Bayes-risk figure of merit, the variance-reduction reward and its telescoping identity all hold.

Two learned policies were trained and compared. TRPO and CEM land in a statistical tie over $1.00\text{e}+04$ held-out frequencies, at $2.17\text{e}-03$ and $2.14\text{e}-03$ Bayes risk respectively — a 2% difference. Averaged over those frequencies, the posterior variance falls from the prior's $8.33\text{e}-02$ to about $2.10\text{e}-03$ for both policies and flattens out well before the end of the episode.

References

D. Fiderer, J. Schuff, and D. Braun, *Neural-network heuristics for adaptive Bayesian quantum estimation*, PRX Quantum **2**, 020303 (2021); preprint dated 2020-03-05.